



CLOUD SERVICE PROVIDERS

AWS • Google Cloud •
Microsoft Azure

Cloud Service Providers:

AWS vs GCP vs Azure

Detailed, practical and interactive comparison of core cloud web services for B.Tech students.

7 CORE AREAS

Compute • Storage

COMPARE

Service-to-service

LEARN BY DOING

Quizzes • Scenarios

By the end, students should be able to map equivalent cloud services and select services for a simple web application.

1 • IDENTIFY

Recognize the major cloud service categories and the role of each service in a web application.

2 • MAP

Relate AWS, Google Cloud and Azure services that solve similar problems.

3 • EXPLAIN

Describe what happens when a request moves from a user to compute, storage/database and monitoring.

4 • CHOOSE

Given a scenario, identify the cloud service category that fits the requirement.

5 • COMPARE

Distinguish VM-based, managed, serverless and API-oriented approaches.

Interactive rule: don't memorize names first. Start with the problem → identify the service category → then map the provider-specific name.

The same application requirement can be implemented with different provider-specific services.

AWS

Amazon Web Services

Strength in this course:

- EC2 for virtual machines
- S3 for object storage
- RDS for managed relational DB
- Lambda for serverless functions
- API Gateway + CloudWatch

Think: a very broad service catalog.

GOOGLE CLOUD

Google Cloud Platform (GCP)

Core examples:

- Compute Engine for VMs
- Cloud Storage for objects
- Cloud SQL for relational DB
- Cloud Run / Cloud Functions
- API Gateway + Cloud Monitoring

Think: cloud infrastructure plus strong data/application tooling.

MICROSOFT AZURE

Microsoft Azure

Core examples:

- Azure Virtual Machines
- Blob Storage
- Azure SQL Database
- Azure Functions
- API Management + Azure Monitor

Think: cloud services integrated with the Microsoft ecosystem.

Interactive prompt: If all three can host a web application, what changes when you move from one provider to another?

Learn the category first, then remember the provider-specific service name.

Cloud capability	AWS	Google Cloud	Microsoft Azure
Virtual machines	EC2	Compute Engine	Virtual Machines
Object storage	S3	Cloud Storage	Blob Storage
Managed relational DB	RDS	Cloud SQL	Azure SQL Database
Cloud networking	VPC	VPC	Virtual Network (VNet)
Serverless functions	Lambda	Cloud Functions	Azure Functions
API management	API Gateway	API Gateway	API Management
Monitoring	CloudWatch	Cloud Monitoring	Azure Monitor

MEMORY TRICK

AWS: EC2/S3/RDS/Lambda • GCP: Compute/Storage/SQL/Functions • Azure: VM/Blob/SQL/Functions

1. Compute Services

Compute is where your application code actually runs.

AWS • EC2

Virtual machines with selectable CPU, memory, storage and networking. You manage the OS and installed software.

GCP • Compute Engine

Configurable VM instances. Useful when you need OS-level control or a conventional server environment.

AZURE • Virtual Machines

Cloud VMs for Windows or Linux workloads with configurable compute, disks and networking.

WHEN TO USE A VM

- Need OS-level control
- Run existing server software
- Long-running or custom workloads

VM RESPONSIBILITIES

- Choose instance size
- Configure OS and software
- Patch and secure the machine

THINK

A college website suddenly gets 10× traffic. Would you manually create more VMs, or use a managed scaling mechanism?

Key concept: Cloud compute is not “a computer on the internet” only—it is programmable, scalable infrastructure.

Object storage is a foundational cloud web service for files, images, videos, backups and static assets.

Provider	Primary object storage	Typical web use	Mental model
AWS	Amazon S3	Images, uploads, backups, static assets	Bucket → objects
Google Cloud	Cloud Storage	Files, datasets, static content	Bucket → objects
Azure	Blob Storage	Media, documents, backups, data	Storage account → containers → blobs

OBJECT ≠ DATABASE

Object storage stores files/objects. It is not normally where you keep rows and relationships for an application.

KEY IDEAS

Buckets/containers, object names, access control, lifecycle rules, durability and retrieval.

CLASSROOM SCENARIO

A student portal accepts profile photos. Store the photo in object storage and keep its URL/reference in the database.

Quick check: Where should a 2 MB JPG uploaded by a student normally go—object storage or relational database?

Managed databases remove much of the infrastructure work from the application developer.

Provider	Managed relational option	Good fit	Student takeaway
AWS	Amazon RDS	Web apps needing SQL databases	AWS manages much of the DB infrastructure
Google Cloud	Cloud SQL	Managed MySQL/PostgreSQL/SQL Server workloads	Connect application → managed SQL
Azure	Azure SQL Database	Cloud-native relational applications	Managed SQL with Azure integration

RELATIONAL

Tables + rows + relationships + SQL.
Examples: students, courses, registrations, fees.

MANAGED

The provider handles many operational tasks such as provisioning, backups and infrastructure maintenance; exact features vary by service.

DESIGN QUESTION

If 50,000 users access a student portal, what must scale: compute, database, storage—or all three?

Remember: a cloud database is still a database. SQL concepts—tables, keys, indexes, queries—remain important.

Networking controls how users, servers and cloud services communicate.

AWS NETWORKING

VPC

- Subnets
- Route tables
- Security groups
- Internet/NAT gateways
- Load balancing via Elastic Load Balancing

Goal: define and protect the network boundary.

GOOGLE CLOUD NETWORKING

VPC

- Subnets
- Routes
- Firewall rules
- Cloud Load Balancing
- Cloud DNS

Goal: connect resources and control traffic.

AZURE NETWORKING

Virtual Network (VNet)

- Subnets
- Network security controls
- Load Balancer / Application Gateway
- DNS services

Goal: create isolated, controlled connectivity.

INTERACTIVE: TRACE THE REQUEST

**Student browser → DNS → Load Balancer → Application/VM → Database
↘ Object Storage**

Ask: Which component decides where traffic goes? Which component stores the uploaded image?

5. Serverless Services

Run application logic without managing a traditional always-on server.

Provider	Function service	Container/serverless option	Typical trigger
AWS	Lambda	—	HTTP/API, storage event, queue, schedule
Google Cloud	Cloud Functions	Cloud Run	HTTP request, event, Pub/Sub/event sources
Azure	Azure Functions	Container Apps	HTTP, timer, queue/event sources

WHY SERVERLESS?

No need to manage a VM for every small function. The platform handles execution infrastructure and scaling.

GOOD USE CASES

Image processing, form handlers, scheduled jobs, lightweight APIs, event-driven tasks.

THINK DIFFERENTLY

Instead of “Which server should I run?”, ask “What event should trigger my code?”

Mini-poll: A function that runs only when a student uploads a file—VM or serverless function?

APIs expose functionality; messaging services let components communicate without being tightly coupled.

Capability	AWS	Google Cloud	Azure
API management	API Gateway	API Gateway	API Management
Queue / messaging	SQS / SNS	Pub/Sub	Service Bus / Event Grid
Typical role	Expose APIs; queue or publish events	Publish/subscribe and event delivery	Enterprise messaging and event routing

API GATEWAY IDEA

Client → API Gateway → backend service

Gateway can provide a controlled entry point for APIs, with capabilities such as routing, authentication/integration patterns and monitoring depending on provider.

QUEUE IDEA

Producer → Queue → Consumer

The producer does not need the consumer to be available at the exact same moment. This helps decouple workloads.

Interactive challenge: An online exam submission should be processed even if the result-generation service is temporarily busy. What pattern helps?

7. Monitoring & Observability

A deployed application is not finished until you can observe whether it is healthy.

Provider	Monitoring service	What students should observe
AWS	CloudWatch	Metrics, logs, dashboards and alarms
Google Cloud	Cloud Monitoring / Cloud Logging	Metrics, dashboards, alerting and logs
Azure	Azure Monitor	Metrics, logs, alerts and application/resource insights

METRICS

CPU utilization, request count, latency, error rate, memory and other service-specific measurements.

LOGS

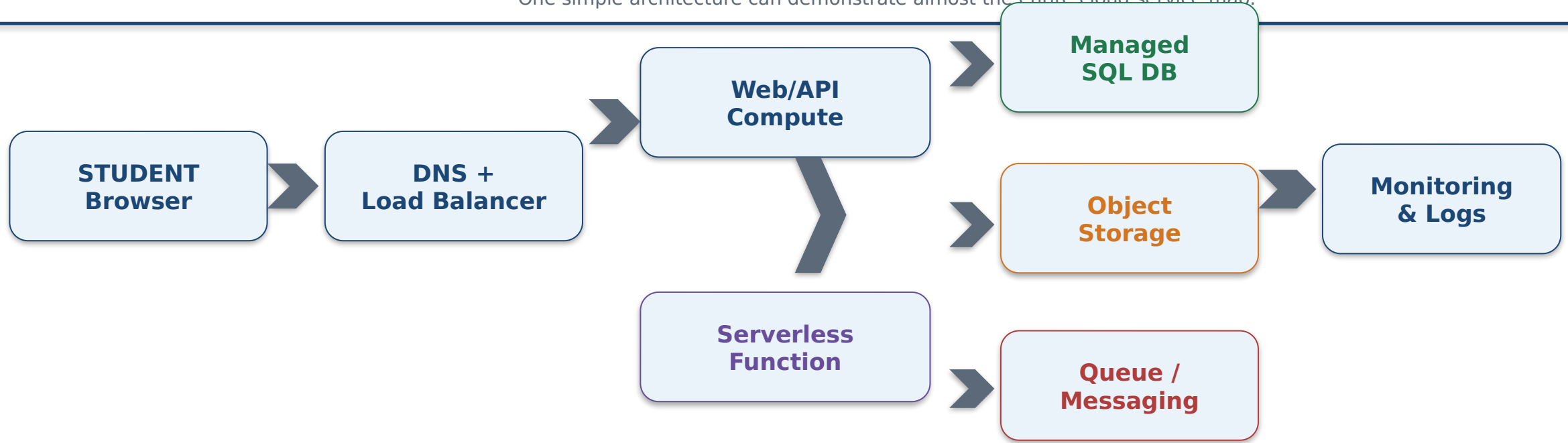
Timestamped application/system records used to understand what happened during a request or failure.

ALERTS

Rules can notify a team when a monitored condition crosses a threshold or meets a defined condition.

Scenario: response time suddenly increases. Which evidence would you inspect first—metrics, logs, or both?

One simple architecture can demonstrate almost the entire cloud service map.



CLASS ACTIVITY: Map every box to AWS, GCP and Azure service names.

Example: Compute → EC2 / Compute Engine / Azure VM
 Storage → S3 / Cloud Storage / Blob Storage
 Database → RDS / Cloud SQL / Azure SQL Database

Work in pairs. Identify the service category first, then name the provider-specific service.

A • REQUIREMENT

Store student profile images

REVEAL → Object
Storage

B • REQUIREMENT

Run a Linux web server with full OS control

REVEAL → Virtual
Machine

C • REQUIREMENT

Store student records in SQL tables

REVEAL → Managed
Relational Database

D • REQUIREMENT

Run code only when an upload event occurs

REVEAL →
Serverless Function

E • REQUIREMENT

Expose /students REST endpoints

REVEAL → API
Management /
Gateway

F • REQUIREMENT

Alert when error rate rises

REVEAL →
Monitoring &
Alerting

Extension: Repeat the exercise for AWS, GCP and Azure. Can you name the equivalent service in all three?

Choose services for a registration system handling forms, photos, database records and notifications.

REQUIREMENTS

- Students submit registration forms.
- Profile/photo files must be uploaded.
- Registration records need SQL queries.
- Submission processing may happen asynchronously.
- System should expose REST APIs.
- Faculty should see health/error information.

PICK A SERVICE CATEGORY

Compute

Web/API application

Object Storage

Photos/files

Relational DB

Registration data

Messaging

Async processing

API Gateway

REST entry point

Monitoring

Metrics + logs + alerts

Discussion: There is more than one valid architecture. Justify your choices using requirements, not provider popularity.

Quick Check: 6 Questions

Use this as a classroom quiz or live discussion. Reveal answers after students vote.

Q1

Which service stores files as objects?

Answer: S3 / Cloud Storage / Blob Storage

Q2

Which AWS service provides VMs?

Answer: EC2

Q3

Which Azure service is the managed relational SQL option?

Answer: Azure SQL Database

Q4

Which GCP service can run containerized web services without VM management?

Answer: Cloud Run

Q5

What service category provides a controlled API entry point?

Answer: API Gateway / API Management

Q6

Which category helps detect high latency and errors?

Answer: Monitoring / Observability

Scoring suggestion: 1 point each • 5-6 = strong grasp • 3-4 = revise mappings • 0-2 = revisit the service categories

Students should finish by deploying one small web application and identifying the equivalent services across providers.

01

CREATE

Launch a basic compute/serverless workload

02

STORE

Upload an image/file to object storage

03

CONNECT

Connect the application to a managed SQL database

04

EXPOSE

Create an API endpoint

05

OBSERVE

Enable logs, metrics and an alert

06

COMPARE

Write the AWS ↔ GCP ↔ Azure service mapping

Final learning outcome: students should be able to look at a web application requirement and translate it into cloud service categories—then map those categories to AWS, GCP or Azure.